

LESSON 5-5 • HOMEWORK PACKET • SHEET 1 OF 2

Volume of Rectangular Prisms

Maryland Standard 6.G.A.2 • Learning target – I can find the volume of a rectangular prism by multiplying length, width, and height.

REMEMBER

- Volume** counts the unit cubes that fill a solid. Use **cubic units** (cm^3 , in^3 , ft^3).
- $V = l \times w \times h$, or $V = B \times h$, where B is the area of the base.

WORKED EXAMPLE

A box is 7 cm long, 3 cm wide, and 5 cm tall.

Step 1 One layer of cubes: $B = 7 \times 3 = 21$

Step 2 Stack 5 layers: $21 \times 5 = 105$

Answer 105 cm^3

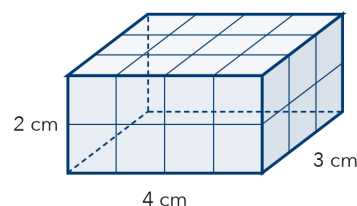
PRACTICE

show your work • label units

1. Each small cube is 1 cm^3 . Find the volume of the prism.

Show your work.

Answer _____ cm^3



2. Find the volume of each rectangular prism.

a. $8 \text{ in} \times 5 \text{ in} \times 2 \text{ in}$ _____ in^3

b. $10 \text{ cm} \times 10 \text{ cm} \times 4 \text{ cm}$ _____ cm^3

c. $6 \text{ ft} \times 6 \text{ ft} \times 6 \text{ ft}$ _____ ft^3

d. $B = 45 \text{ m}^2$, $h = 7 \text{ m}$ _____ m^3

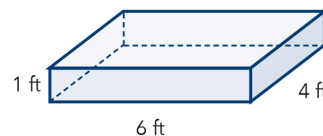
e. $12 \text{ m} \times 3 \text{ m} \times 5 \text{ m}$ _____ m^3

f. $B = 32 \text{ in}^2$, $h = 9 \text{ in}$ _____ in^3

3. **Sandbox.** A sandbox is 6 ft long, 4 ft wide, and 1 ft deep. One bag of sand fills 2 ft^3 . How many bags fill the sandbox?

Show your work.

Answer _____ bags



4. **Fish Tank.** A fish tank is 20 in long and 10 in wide. It holds $3,000 \text{ in}^3$ of water when full. How tall is the tank?

Answer _____ in

5. **Compare.** Box A is $6 \text{ in} \times 4 \text{ in} \times 2 \text{ in}$. Box B is $3 \text{ in} \times 4 \text{ in} \times 4 \text{ in}$. Mika says Box A holds more because it is longer. Is she right? Explain.

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____

AT HOME Find a box at home. Measure or estimate its length, width, and height, and find its volume.

LESSON 5-5 • HOMEWORK PACKET • SHEET 2 OF 2**Missing Fractional Edges**

Maryland Standard 6.G.A.2 • Learning target – I can find a missing edge of a prism and find volumes with fractional edge lengths.

REMEMBER

- $V = l \times w \times h$ works with fractions too. Write mixed numbers as fractions first.
- Missing edge: **divide** the volume by the product of the other two edges.

WORKED EXAMPLE

A box is $2\frac{1}{2}$ ft $\times$ 2 ft $\times$ $1\frac{1}{2}$ ft. Find its volume.

Step 1 Write as fractions: $5/2 \times 2/1 \times 3/2$

Step 2 Multiply: $(5 \times 2 \times 3)/(2 \times 1 \times 2) = 30/4$

Step 3 Simplify: $30/4 = 7\frac{1}{2}$

Answer $7\frac{1}{2}$ ft³

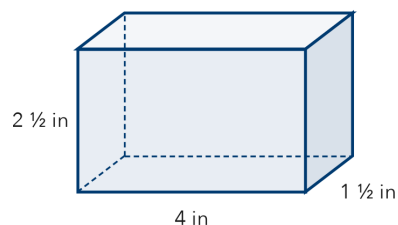
PRACTICE

show your work • label units

1. Find the volume of the prism.

Show your work.

Answer _____ in³



2. Find each volume.

a. $\frac{1}{2}$ ft $\times$ 6 ft $\times$ 8 ft _____ ft³

b. $3\frac{1}{2}$ m $\times$ 2 m $\times$ 4 m _____ m³

c. $2\frac{1}{2}$ cm $\times$ 4 cm $\times$ 3 cm _____ cm³

d. $1\frac{1}{2}$ in $\times$ $1\frac{1}{2}$ in $\times$ 4 in _____ in³

e. 4 ft $\times$ $\frac{1}{2}$ ft $\times$ 5 ft _____ ft³

f. 6 cm $\times$ $1\frac{1}{2}$ cm $\times$ 3 cm _____ cm³

3. **Planter.** A planter box is 3 ft long and $1\frac{1}{2}$ ft wide. It holds 9 ft³ of soil when full. How deep is it?

Answer _____ ft

4. **Small Cubes.** A box is 2 in long, 1 in wide, and $1\frac{1}{2}$ in tall. It is packed with cubes that are $\frac{1}{2}$ in on each edge.

a. How many cubes fill the box? _____

b. Each cube is $\frac{1}{8}$ in³. What is the volume of the box? _____ in³

TEST PRACTICE

choose the best answer

5. A box is 5 in long, 2 in wide, and $1\frac{1}{2}$ in tall. What is its volume?

A $8\frac{1}{2}$ in³

B 10 in³

C 15 in²

D 15 in³

6. A prism has a volume of 96 ft³. Its base is 4 ft by 6 ft. What is its height?

A 4 ft

B 16 ft

C 24 ft

D 576 ft

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about # _____

AT HOME Find a box at home with a fraction edge (like $2\frac{1}{2}$ in). Estimate its volume.